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Inquiry on InBody Measurement Accuracy

Common question asked by the customer is, Is the InBody result correct? The right answers as shown below.

1. Response

1) Standards on PBF Values Between Different Models

- **Height and Weight values must be same for continuous measurement.**
- ➢ ±1% when continuously measured on the same product
- ➢ ±2% when continuously measured between different product but same model
- ➢ ±3% when continuously measured between different models

- Reasons for Difference in the PBF % Values

① Environmental Factor: ±1% considered due to surrounding temperature, humidity, slight movements, micro substances and etc.

② Professional Products – Inquiry on Accuracy Between Professional Products

Difference Between Professional Products	*Error Range: [Continuous Measurement: ±3%] Error range of 3% is due to 1% from environmental factors, 1% from the range of frequencies used, and 1% from the status of a human body. For Professional Products, the range of frequencies are from minimum 2 frequencies (20, 100 kHz) to 6 frequencies (1, 5, 50, 250, 500, 1000 kHz) depending on the model. Accurate measurement of the Body Fat and Muscle Mass can be analyzed with two frequencies but analysis on the ICW and ECW ratio is more precise as the range of frequencies increase even for patients with edema. Therefore, variety of parameters of the human body composition can be displayed as the number of frequencies increase.

③ Professional Products – Inquiry on Accuracy with Home Use Products

Professional Products	*Error Range: [Continuous Measurement: ±3%] 8 Point Tactile measurement method is same between Professional Products and Home Use Products. Error range of 3% is due to 1% from environmental factors, 1% from the range of frequencies used, and 1% from the status of a human body. InBody Dial uses total number of 2 frequencies which are 20, 100 kHz. For Professional Products, the range of frequencies start from 2frequencies to 6 frequencies.
InBody Dial	
Professional Products	* Error Range : [Range of error: X] The principle of analyzing body composition in InBody Band and Professional Products are same which is BIA. Professional InBody measures the human body in 5 segments, including the lower body. However, InBody Band only measures the body composition of the upper body and estimates the body composition of the lower body. For those who either have developed upper body or the lower body will have difference in the measurement between the InBody Band and the Professional Products therefore there is no fixed error range. The major purpose of the InBody Band is to track the daily activity. Therefore, it is recommended to be used to monitor the trend in the progress of the body composition.
InBody Band	

④ Home Product – Inquiry on Accuracy Home Use Products

Between InBody Dial	* Error Range: [Continuous Measurement on Same Product: Less than ±1%] [Continous Measurement Between Same Model but Different Product: Less than ±2%] Error range of 2% is due to 1% from environmental factors, and 1% from the status of a human body.
Between InBody Band	* Error Range : [Range of error: X] The principle of analyzing body composition in InBody Band and InBody Dial are same which is BIA. InBody Dial measures the human body in 5 segments, including the lower body. However, InBody Band only measures the body composition of the upper body and estimates the body composition of the lower body. For those who either have developed upper body or the lower body will have difference in the measurement between the InBody Band and the InBody Dial therefore there is no fixed error range. The major purpose of the InBody Band is to track the daily activity. Therefore, it is recommended to be used to monitor the trend in the progress of the body composition.

2. Procedure in Testing Accuracy and Reproducibility

1) Tools Required: Calibration Unit

2) Procedure

① Connect the Calibration Unit to the product.

② Enter the Test Mode, press "4. Imp. Calib." then press "1. Start" to perform Impedance Calibration.

③ Print the QC Report from the menu "QC Print".

④ Check the Impedance Values under Accuracy and Reproducibility of the R2 Range in "Unit Calibration Data".

3) Checking Measured Values (Unit: Ω)

Segment	Standard Value (QC ORG Data)	Standard Measurement Value
Right/Left arm	300	297.9 ~ 302.1 (±0.7%)
Right/Left leg	250	248.25 ~ 251.751 (±0.7%)
Body	25	24.25 ~ 25.75 (±3.0%)

4) RC Unit test

① Tools Required: RC Unit (Human Jig)

②Procedure

(1) Connect the RC unit (Human Jig) to the product.

(2) Change the Weight Measurement to Manual Mode from the Setup.

*The Consumer Devices (Dial) do not need a test weight but only Calibration Unit H20/H30 must be connected to the InBody Dial. Change the Calibration Unit to RC Mode and enter the hidden menu on the device > Imp Cal > Conti.WT menu to run a automatic RC Jig test. (Please refer to the H30 CS Training Manual for more details.)

(3) Place an object of which is over 5 kg on the Foot Case.

(4) Set the gender (male) / age (30 years) / Weight (70 kg) / Height (170 cm) and perform the InBody Test.

(5) Compare the PBF values with the standard provided in the chart below.

• Standards on PBF using RC Unit (Human Jig) in Different Models

(Calibration Unit 970,BWA2.0: RC1, Calibration Unit 380/580/S Series: RC2)

Model	Standard PBF	Min. (-1%)	Max. (+1%)
InBody270	17.0%	16.0%	18.0%
InBody470	29.0%	28.0%	30.0%
InBody370	28.0%	29.0%	30.0%
InBody370S	29.5%	28.5%	30.5%
InBody570	35%	34%	36%
InBody770	29.8%	28.8%	30.8%
InBodyS10	29.5%	28.5%	30.5%

InBodyJ30	29%	28%	30%
InBodyJ50	29%	28%	30%
InBodyH20	12.8% (±3 %)	9.8%	15.8%
InBodyH30	27.0%	26.0%	28.0%
BWA ON	28.0%	27.0%	29.0%
BWA 2.0S	36.4% (±0.8%)	35.6%	37.2%
BWA 2.0_Clamp	40.9% (±0.6%)	40.3%	41.5%
BWA 2.0_Adhesive	39.0% (±0.6%)	38.4%	39.6%
InBody270S	25.5% (±0.8%)	24.7%	26.3%
InBody380(Korea)	26.8% (±0.8%)	26.0%	27.6%
InBody380(Overseas)	21.5% (±0.8%)	20.7%	22.3%
InBody580	26.5% (±0.8%)	25.7%	27.3%
InBody770S	26.5% (±0.8%)	25.7%	27.3%
InBody970S	26.0% (±0.8%)	25.2%	26.8%
InBody470S	25.5% (±0.8%)	24.7%	26.3%

5) Accuracy and Reproducibility of Weight Scale

① Enter the Test Mode, press "4. Load cell Calibration" in 3. NAWI Mode.

② Place 40kg of weight (2 pieces of 20kg counterweights) and check for the accuracy and reproducibility.

(1) Error Range for Accuracy: The difference in the range should be less than ±50 g.

(2) Error Range for Reproducibility: The difference between the weights should be less than ±50 g.